

Unit A

Non-normality of the errors

We suppose that $y_i = x_i^T \beta + \varepsilon_i$, with $\varepsilon_i \stackrel{\text{iid}}{\sim} \mathcal{F}$ such that

$$\mathbb{E}[\varepsilon_i] = 0 \text{ and } \text{var}(\varepsilon_i) = \sigma^2 \mathbf{I}_m.$$

"any distribution, such as Laplace, skew-normal, logistic, etc"

The OLS estimator $\hat{\beta} = (X^T X)^{-1} X^T y$ satisfies

$$\mathbb{E}[\hat{\beta}] = \mathbb{E}[(X^T X)^{-1} X^T y] = (X^T X)^{-1} X^T \mathbb{E}[y] =$$

$$= (X^T X)^{-1} X^T (X\beta + \mathbb{E}[\varepsilon]) = \cancel{(X^T X)^{-1}} \cancel{X^T X} \beta = \beta$$

It's unbiased!

"by assumption, no need to be Gaussian!"

$$\text{var}(\hat{\beta}) = \text{var}((X^T X)^{-1} X^T y) = (X^T X)^{-1} X^T \text{var}(y) [(X^T X)^{-1} X^T]^T$$

↳ Properties of the variance

$$\text{var}(Ay) = A \text{var}(y) A^T.$$

$$\text{var}(y) = \text{var}(X\beta + \varepsilon) = \text{var}(\varepsilon) = \sigma^2 \mathbf{I}_m$$

$$= (X^T X)^{-1} X^T \sigma^2 \mathbf{I}_m X [(X^T X)^{-1}]^T$$

$$= \sigma^2 \cancel{(X^T X)^{-1}} \cancel{X^T X} (X^T X)^{-1}$$

↑ This is symmetric.

$$= \sigma^2 (X^T X)^{-1}$$

can be written as a sum, for suitable weights δ_{ij} .

$$\text{Note that } \hat{\beta}_\delta = (X^T X)^{-1} X^T y_i = \sum_{i=1}^m \delta_{ij} y_i$$

↳ Not surprising that a CLT holds.

Heteroschedasticity of the errors

Suppose now that $y_i = x_i^T \beta + \varepsilon_i$, but $E[\varepsilon] = 0$ and $\text{var}(\varepsilon) = \Sigma$.

diag matrix.

To fix ideas, suppose that

$$\Sigma = \text{diag}(\sigma_1^2, \dots, \sigma_n^2) = \begin{bmatrix} \sigma_1^2 & & 0 \\ & \ddots & \\ 0 & & \sigma_n^2 \end{bmatrix}$$

Consider again the proof of before

$$E[\hat{\beta}] = (X^T X)^{-1} X^T E[y] = (X^T X)^{-1} X^T [X\beta + \underbrace{E[\varepsilon]}_{0!}] = \cancel{(X^T X)^{-1}} \cancel{X^T X} \beta = \beta$$

With similar steps as before

$$\text{var}(\hat{\beta}) = (X^T X)^{-1} X^T \underbrace{\text{var}(y)}_{\substack{= \\ \text{var}(\varepsilon) \\ = \\ \Sigma}} [(X^T X)^{-1} X^T]^T = (X^T X)^{-1} X^T \Sigma X (X^T X)^{-1}$$

Weighted Least Squares

Let us consider again the case of heteroskedastic errors:

$$\text{var}(\varepsilon) = \sigma^2 \Omega^{-1} \Leftrightarrow \text{var}(y_i) = \sigma_i^2 = \sigma^2 / w_i,$$

$$\Omega = \text{diag}(w_1, \dots, w_m). \quad \text{This is now treated as known}$$

Let us define

$$y^* = \Omega^{1/2} y, \quad X^* = \Omega^{1/2} X,$$

that is $y_i^* = \sqrt{w_i} y_i$, $x_{ij}^* = \sqrt{w_i} x_{ij}$. Then:

$$\mathbb{E}[y^*] = \mathbb{E}[\Omega^{1/2} y] = \Omega^{1/2} \mathbb{E}[y] = \Omega^{1/2} X\beta = X^*\beta.$$

$$\begin{aligned} \text{var}(y^*) &= \text{var}(\Omega^{1/2} y) = \Omega^{1/2} \text{var}(y) \Omega^{1/2} = \sigma^2 \Omega^{1/2} \Omega^{-1} \Omega^{1/2} = \\ &\quad \text{"} \quad \text{var}(\varepsilon) \quad = \sigma^2 \cancel{\Omega^{1/2}} \Omega^{-1} \cancel{\Omega^{1/2}} = \\ &\quad \text{"} \quad \sigma^2 \Omega^{-1/2} = \sigma^2 I_m. \end{aligned}$$

$\Rightarrow$ The OLS assumptions are valid for $y^* = X^*\beta + \varepsilon^*$. Hence

$$D_{\text{wls}}(\beta) = (y^* - X^*\beta)^T (y^* - X^*\beta)$$

$$= (\Omega^{1/2} y - \Omega^{1/2} X\beta)^T (\Omega^{1/2} y - \Omega^{1/2} X\beta)$$

$$\begin{aligned} &= (y - X\beta)^T \underbrace{\Omega^{1/2} \Omega^{1/2}}_{\Omega} (y - X\beta) = (y - X\beta)^T \Omega (y - X\beta) \\ &= \sum_{i=1}^m w_i (y_i - x_i^T \beta)^2 \end{aligned}$$

$$\Rightarrow \hat{\beta}_{\text{wls}} = \underset{\beta \in \mathbb{R}^p}{\text{argmin}} D_{\text{wls}}(\beta) = \underset{\beta \in \mathbb{R}^p}{\text{argmin}} \sum_{i=1}^m w_i (y_i - x_i^T \beta)^2$$

$$= [(X^*)^T X^*]^{-1} (X^*)^T y^* = [(\Omega^{1/2} X)^T \Omega^{1/2} X]^{-1} (\Omega^{1/2} X)^T \Omega^{1/2} y$$

$$= [X^T \Omega^{\frac{1}{2}} \Omega^{\frac{1}{2}} X]^{-1} X^T \Omega^{\frac{1}{2}} \Omega^{\frac{1}{2}} y = (X^T \Omega X)^{-1} X^T \Omega y.$$

Such an estimator is unbiased and efficient (BLUE), with

$$E[\hat{\beta}_{wls}] = E[(X^T \Omega X)^{-1} X^T \Omega y] = (X^T \Omega X)^{-1} X^T \Omega E[y] = \cancel{(X^T \Omega X)^{-1} X^T \Omega X} \beta = \beta.$$

Moreover

$$\text{var}(\hat{\beta}_{wls}) = \text{var}((X^T \Omega X)^{-1} X^T \Omega y) = (X^T \Omega X)^{-1} X^T \Omega \underset{\sigma^2 \Omega^{-1}}{\text{var}(y)} [(X^T \Omega X)^{-1} X^T \Omega]^T$$

$$= \sigma^2 (X^T \Omega X)^{-1} \cancel{X^T \Omega} \Omega^{-1} \Omega X (X^T \Omega X)^{-1}$$

$$= \sigma^2 (X^T \Omega X)^{-1} \cancel{X^T \Omega X} (X^T \Omega X)^{-1} = \sigma^2 (X^T \Omega X)^{-1}.$$

Box-Cox transformation

The Box-Cox transformation is

$$g_\lambda(y) = \frac{y^\lambda - 1}{\lambda}, \quad \lambda \neq 0.$$

Note that $\lim_{\lambda \rightarrow 0} \frac{y^\lambda - 1}{\lambda} = \log(y)$. We consider the transformed data

$$g_\lambda(y_i) = x_i^T \beta + \varepsilon_i, \quad \varepsilon_i \sim N(0, \sigma^2)$$

therefore $z_\lambda = (g_\lambda(y_1), \dots, g_\lambda(y_m)) \sim N_m(X\beta, \sigma^2 I_m)$,

where the joint density is

$$f_z(z_\lambda) = \frac{1}{(2\pi\sigma^2)^{m/2}} \exp\left\{-\frac{1}{2\sigma^2}(z_\lambda - X\beta)^T(z_\lambda - X\beta)\right\}$$

We are interested in the joint density of the original data $y_i = g_\lambda^{-1}(z_i)$

$$f_y(y) = f_z(g_\lambda(y_1), \dots, g_\lambda(y_m)) \prod_{i=1}^m \left| \frac{\partial g_\lambda(y_i)}{\partial y_i} \right|$$

Jacobian term $\left| \frac{\partial g_\lambda(y_i)}{\partial y_i} \right| = y_i^{\lambda-1}$

The log-likelihood becomes

$$\ell(\beta, \sigma^2, \lambda) = \log f_y(y) =$$

$$= \log \left[\frac{1}{(2\pi\sigma^2)^{m/2}} \exp\left\{-\frac{1}{2\sigma^2}(z_\lambda - X\beta)^T(z_\lambda - X\beta)\right\} \right] + \log \left(\prod_{i=1}^m y_i^{\lambda-1} \right) + c$$

$$= -\frac{n}{2} \log(\sigma^2) - \frac{n}{2} \log(2\pi) - \frac{1}{2\sigma^2} (z_\lambda - X\beta)^T (z_\lambda - X\beta) + (\lambda-1) \sum_{i=1}^n \log(y_i) + c$$

For fixed values of λ , the maximizer of $\hat{\beta}$, $\hat{\sigma}^2$ are the usual estimates:

$$\hat{\beta}_\lambda = (X^T X)^{-1} X^T z_\lambda, \quad \hat{\sigma}^2 = \frac{1}{n} (z_\lambda - X\hat{\beta}_\lambda)^T (z_\lambda - X\hat{\beta}_\lambda)$$

$$\Rightarrow \ell_p(\lambda) = \ell(\hat{\beta}_\lambda, \hat{\sigma}_\lambda^2, \lambda) =$$

$$= -\frac{n}{2} \log(\hat{\sigma}_\lambda^2) - \frac{1}{2\hat{\sigma}_\lambda^2} \cdot n \hat{\sigma}_\lambda^2 + (\lambda-1) \sum_{i=1}^n \log(y_i) + c$$

↓ The entire term is a constant

This quantity should be maximized numerically.

Variance-stabilizing transformations. Poisson case

Let $y_i \stackrel{\text{ind}}{\sim} \text{Poisson}(\mu_i)$, with $\mathbb{E}[y_i] = \mu_i = \text{var}(y_i)$. Note that

$$y_i \sim N(\mu_i, \mu_i) \text{ for large value of } \mu_i$$

The probabilists would say that

$$\frac{y_i - \mu_i}{\sqrt{\mu_i}} \xrightarrow{d} Z, \quad Z \sim N(0, 1) \text{ as } \mu_i \rightarrow \infty.$$

However, the data are heteroskedastic. What about considering a transformation

$g(y_i)$? Note that, informally

Linear approximation

$$g(y_i) - g(\mu_i) \approx (y_i - \mu_i) g'(\mu_i) \Rightarrow \text{var}(g(y_i)) \approx g'(\mu_i)^2 \underbrace{\text{var}(y_i)}_{\mu_i}.$$

This is another way to look at the delta method:

$$\frac{g(y_i) - g(\mu_i)}{\sqrt{\mu_i}} \xrightarrow{d} Z, \quad Z \sim N(0, g'(\mu_i)^2)$$

$$\Rightarrow g(y_i) \sim N(g(\mu_i), \mu_i g'(\mu_i)^2)$$

Now, the simple idea is to pick $g(y_i)$ such that $\mu_i g'(\mu_i)^2 = \text{"constant"}$.

This choice is $g(y_i) = \sqrt{y_i}$. Indeed

$$\text{var}(\sqrt{y_i}) \approx \mu_i \left| \frac{\partial \sqrt{y}}{\partial y} \right|_{\mu_i}^2 = \mu_i \left(\frac{1}{2\sqrt{\mu_i}} \right)^2 = \frac{1}{4} \frac{\mu_i}{\mu_i} = \frac{1}{4}$$

Variance-stabilizing transformation - Binomial case

Let $y_i \sim \text{Binomial}(m_i, \tilde{\pi}_i)$, with success probability $\tilde{\pi}_i = g(x_i, \beta)$. For large m_i , we get

$$y_i \sim N(m_i \tilde{\pi}_i, m_i \tilde{\pi}_i (1 - \tilde{\pi}_i))$$

Reasoning as before, we obtain that

$$g(y_i) - g(\mu_i) \approx (y_i - \mu_i) g'(y_i) \Rightarrow \text{var}(g(y_i)) \approx m_i \tilde{\pi}_i (1 - \tilde{\pi}_i) g'(\mu_i)^2.$$

with $\mu_i = m_i \cdot \tilde{\pi}_i$. The special choice gives an approximately constant variance

$$g_{m_i}(y) = \sqrt{m_i} \arcsin\left(\frac{2y}{m_i} - 1\right)$$

Depends on m_i ...

The variance is in fact equal to

$$\text{var}(g_{m_i}(y_i)) \approx m_i \tilde{\pi}_i (1 - \tilde{\pi}_i) \left[\sqrt{m_i} \cdot \frac{1}{\sqrt{1 - \left(\frac{2\tilde{\pi}_i m_i}{m_i} - 1\right)^2}} \cdot \frac{2}{m_i} \right]^2$$

$$\frac{\partial}{\partial x} \arcsin(x) = \frac{1}{\sqrt{1-x^2}}$$

$$= \cancel{m_i} \tilde{\pi}_i (1 - \tilde{\pi}_i) \cdot \cancel{m_i} \cdot \frac{4}{\cancel{m_i^2}} \cdot \frac{1}{1 - (2\tilde{\pi}_i - 1)^2}$$

$$= \frac{4\tilde{\pi}_i(1-\tilde{\pi}_i)}{1 - (4\tilde{\pi}_i^2 - 4\tilde{\pi}_i + 1)} = \frac{4\tilde{\pi}_i(1-\tilde{\pi}_i)}{4\tilde{\pi}_i - 4\tilde{\pi}_i^2} = \frac{4}{4} \frac{\tilde{\pi}_i(1-\tilde{\pi}_i)}{\tilde{\pi}_i(1-\tilde{\pi}_i)} = 1.$$